Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/13/2024
PI Name:	Weber, Richard Ph.D.		
Project Title:	The Origin of Fragility in High-Temperature Oxide Liquids		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Materials Development Inc.		
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City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Japanese Space Agency (JAXA) AO for Electrostatic Levitation Furnace (ELF) experiments on ISS
Start Date:	08/15/2019	**End Date:**	09/30/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2025 per NSSC information (Ed., 10/9/24) NOTE: End date changed to 9/30/2024 per NSSC information (Ed., 3/5/24) NOTE: End date changed to 9/30/2023 per NSSC information (Ed., 6/20/23) NOTE: End date changed to 9/30/2022 per M. Sansoucie/MSFC (Ed., 1/28/21) June 2019: Note project had preliminary work performed Feb-August 2019		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1288 ; 80NSSC19K0518		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is the United States (US) portion of an international Japan Aerospace Exploration Agency (JAXA) program to investigate the properties and structure of molten and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS) and ground-based synchrotron x-ray and neutron studies. Four prototype compositions have been levitated and melted on ISS. Measurements of thermophysical properties, density, thermal expansion coefficient, viscosity and surface tension were made over a wide temperature range. Samples, including glasses, were recovered for ground based analysis and comparison with materials made on Earth. The work underpins development of a materials property-structure data set that can help in development of new materials and in meeting requirements of material utilization and processing in space. Additional samples are being prepared for measurements on ISS in 2023. Work is being published to provide data to the broader scientific community. The project team brings together the skills and experience needed to implement a complex research program. Reference: [1] S. Kohara, et al., “The origin of fragility in high-temperature oxide liquids - towards fabrication of novel non-equilibrium oxides,” JAXA flight project funded in 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Developing an advanced understanding of the properties and structure of liquids that can be transformed into glasses is a high value enterprise that benefits the $300B global glass materials industry. Specific advances expected from this research will impact development of functional glasses by providing high accuracy thermophysical property and atomic structure data that can be used to optimize models used to design new materials from first principles. The titanate liquids being studied have high value applications. The compositions also relate to lunar mare regolith that contains relatively large amounts of titanium oxide. The work has led to discovery a new type of glass with highly coordinated titanium ions. Work is/has been published and a US patent on an optical device glass was filed.
Task Progress:	This project is part of a collaborative international Japan Aerospace Exploration Agency (JAXA) / NASA program to investigate: (i) properties of liquid phase and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS) [1,2]; and (ii) atomic structure using ground-based synchrotron and neutron studies. We, the NASA investigators, are supplying samples for experiments using the ISS ELF to provide some of the first results from the instrument for US investigators. Samples were launched on SpX-20 in spring of 2020, on SpX-22 in June 2021, and on SpX-27 in March 2023. A holder was run in February 2024 with >80% success rate in melting levitated samples. In addition to several cited publications, the results are establishing a new class of optical glass materials and have led to new technology that is being patented. 1. Status: The project team includes Dr. Stephen Wilke, Mr. Abdulrahman Al-Rubkhi, and Mr. Jared Rafferty from Materials Development, Inc. (MDI); Dr. Shinji Kohara, National Institute for Materials Science, Japan; and Dr. Takehiko Ishikawa, Japanese Aerospace Exploration Agency, Japan. The collaborators from Japan are not funded from this grant. 2. Technical Activities: Activities during the reporting period included: (i) Running ISS ELF experiments on samples launched in mid-2023, (ii) Analyzing data from flight experiments, (iii) Performing ground-based measurements on samples recovered from the earlier flight experiments, and (iv) Publishing/presenting results. (i) Three sets of flight experiments have been performed. The first was in August 2021, the second in December 2021 through March 2022, and the third in February 2024. During the February 2024 experiments, Richard Weber and Stephen Wilke traveled to the JAXA facility in Tsukuba to participate in the work. This single visit to JAXA was important to develop a full understanding of the experimental work. During these experiments, 16 of 19 samples were levitated and melted in ELF. A total of 30 samples covering 7 different compositions were investigated. For four compositions, multiple samples were stably levitated and fully melted. Drop size and oscillation power spectra were measured, and samples (glass for one composition) were recovered. Samples of the RT-3 composition for experiments in February 2024 were larger than previously used: 2.2 instead of 2.0 mm diameter. All samples used in the prior work formed glass when they were cooled in the ELF. This means that the critical cooling rate for glass formation was achieved. The larger samples cooled at a slower rate due to their smaller surface area:volume. Since the larger samples also formed glass, the results demonstrate a decrease in the critical cooling rate compared to the same composition samples processed on Earth. Cooling rate data were measured for several samples made in ELF and on the ground using aerodynamic levitation. The cooling rates on the ground resulted from both radiative and forced convective cooling. The stagnant gas in the ELF experiments means that cooling was mainly by radiation with some conduction through the gas. The smallest cooling rate in the ISS experiments was for the 2.2 mm diameter samples. Glasses were made at somewhat smaller cooling rates in space than in the ground-based measurements. This is an important result. It suggests that the critical cooling rate in reduced gravity is slightly smaller than in Earth-based experiments where extensive fluid motion occurs. (ii) Analyses were performed on the video images, drop oscillation power spectra, and temperature data to obtain the temperature-dependent properties of the liquids. During the early part of the work, considerable effort was put into developing procedures for the analysis and to enable precise synchronization of the temperature and other measured data. A paper that describes the methods has been submitted and is in peer review. Results of density, thermal expansion coefficient, surface tension and viscosity have been published for several compositions. Please see cited references [3,4]. (iii) Ground based experiments and modeling are proceeding in several areas. These include: 1. Measurements of the atomic structure and dynamics of liquids and glasses. 2. Measurements of internal structure of samples using x-ray tomography. 3. Optical property measurements including fluorescence and Raman. 4. Thermodynamic property measurements on samples made on the ground and on ISS. The most detailed atomic structure measurements were made on the RT-3 neodymium titanate composition glasses. These were studied using both x-ray and neutron diffraction, including with isotope substitution. This combination of measurements provided the basis for a detailed and well-constrained structure model for the glass network that has been cited extensively[5]. As the Advanced Photon Source (APS) is currently closed for upgrade, MDI has sent some rare earth titanate composition glass samples to Dr. Shinji Kohara for measurements of the pair distribution functions (pdf) at the SPring-8 synchrotron facility. In collaboration with Professor Collin Wilkinson at Alfred University, molecular dynamics (MD) simulations have been

	developed to investigate the diffusion of different species in the molten RT-3 composition. These simulations use interatomic potentials that were refined to provide good agreement with X-ray scattering data on levitated melts, and the model exhibits viscosities similar to those observed in the ELF measurements. A manuscript is in preparation to publish this study. After completion of atomic structure measurements and release from the neutron source (~August 2023), samples were sent to the Materials of the Universe (MotU) laboratory at Arizona State University. Thermal properties of glasses made on ISS and on the ground were measured using drop solution calorimetry and differential scanning calorimetry. These measurements provide the glass transition temperature and enthalpy of vitrification. Work is being prepared for publication. The optical properties of several glasses have been measured in a collaboration with Dr. Brian Topper at University of New Mexico and Professor Doris Monke at Alfred University. The octahedral titanate network provides a low phonon energy host enabling long fluorescence lifetimes and generation of infrared wavelengths that are typically not possible in hard oxide glasses. In addition, the glasses are strong and stable compared to typical low phonon energy halides. This work has led to four publications [6-9] and a patent filing. Raman spectroscopy measurements made at Alfred University show a composition-dependent trend in the connectivity of the network as a function of rare earth content. This work complements x-ray pdf measurements. (iv) Publishing/presenting results. Several papers have been published based on the ISS experiments. These are listed in the references section. Talks were presented at the ASGSR meeting in Washington, DC (Weber), American Ceramic Society Glass and Optical Materials meeting in Las Vegas, May 2024 (Weber invited, Wilke), 22nd Symposium on Thermophysical Properties in Boulder, CO, June 2024 (Weber) and Non-Crystalline Materials (NCM-15), Cambridge, UK, July 2024 (Weber, Wilke). Additional papers are in preparation. Plans: Plans for the next project period are: (i) Continue data analysis, (ii) Perform post-flight analysis and ground-based measurements on samples from the recent flight experiment, (iii) Continue post-flight sample analysis, (iv) Prepare publications based on the research. 3. References cited [1] https:// [2] R. Weber, S.K. Wilke, and C.J. Benmore, “Containerless Techniques for in-situ X-Ray Measurements on Materials in Extreme Conditions,” J. Phys. Soc. Jpn., 91, 091008 (2022). [3] S.K. Wilke, A. Al-Rubkhi, C. Koyama, T. Ishikawa, H. Oda, B. Topper, E.M. Tsekrekas, D. Möncke, O.L.G. Alderman, V. Menon, J. Rafferty, E. Clark, A.L. Kastengren, C.J. Benmore, J. Ilavsky, J. Neuefeind, S. Kohara, M. SanSoucie, B. Phillips and R. Weber, “Microgravity effects on nonequilibrium melt processing of neodymium titanate: thermophysical properties, atomic structure, glass formation and crystallization,” npj Microgravity, 10:26, (2024). [4] S.K. Wilke, A. Alrubkhi, V. Menon, J. Rafferty, C. Koyama, T. Ishikawa, H. Oda, R. Hyers, R. Bradshaw, A. Kastengren, S. Kohara, M. SanSoucie, B. Phillips, and R. Weber, “Measuring the density, viscosity, and surface tension of molten titanates using electrostatic levitation in microgravity”, Appl. Phys. Lett., in revision. [5] S.K. Wilke, O.L.G. Alderman, C.J. Benmore, J. Neuefeind and R. Weber, “Octahedral oxide glass network in ambient pressure neodymium titanate,” Sci. Rep., 12:8258 (2022), https:// [6] B. Topper, S.K. Wilke, M. Pettes, A. Alrubkhi, V. Menon, A. Neumann, D. Möncke, R. Weber, and A. Mafi, “Mid-infrared luminescence properties of erbium and dysprosium doped lanthanum titanate glasses,” Opt. Mats. Express, 13, 2857 (2023). [7] B. Topper, A. Neumann, A. Mafi, M. Pettes, A. Alrubkhi, S.K. Wilke and R. Weber, “Nonlinear optical properties of lanthanum titanate glasses prepared by levitation melting,” Appl. Phys. Lett., 124, 094104 (2024). [8] B. Topper, A. Neumann, S.K. Wilke, R.E. Youngman, A. Abdulrahman and R. Weber, “Comparing Pr3+ and Nd3+ for deactivating the Er3+: 4I13/2 level in lanthanum titanate glass,” Opt. Mats. Express, 14, 1309 (2024). [9] B. Topper, A. Neumann, S.K. Wilke, A. Alrubkhi, A. Mafi and R. Weber, “Site-selective fluorescence and spectroscopic properties of Yb-doped lanthanum titanate glasses,” Int. J. Glass. Sci., (2024) in press.

Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Wilke SK, Al-Rubkhi A, Koyama C, Ishikawa T, Oda H, Topper B, Tsekrekas EM, Möncke D, Alderman OL, Menon V, Rafferty J. Clark E, Kastengren AL, Benmore CJ, Ilavsky J, Neuefeind J, Kohara S, SanSoucie M, Phillips B, Weber R. "Microgravity effects on nonequilibrium melt processing of neodymium titanate: Thermophysical properties, atomic structure, glass formation and crystallization." npj Microgravity. 2024 Mar 6;10(1):26. https://doi.org/10.1038/s41526-024-00371-x , Mar-2024
Articles in Peer-reviewed Journals	Topper B, Wilke SK, Pettes M, Alrubkhi A, Menon V, Neumann A, Möncke D, Weber R, Mafi A. "Mid-infrared luminescence properties of erbium and dysprosium doped lanthanum titanate glasses." Optical Materials Express. 2023 Oct 1;13(10):2857-68. https://doi.org/10.1364/OME.503121 , Oct-2023
Articles in Peer-reviewed Journals	Topper B, Neumann A, Mafi A, Pettes M, Alrubkhi A, Wilke SK, Weber R. "Nonlinear optical properties of lanthanum titanate glasses prepared by levitation melting." Applied Physics Letters. 2024 Feb 26;124(9). https://doi.org/10.1063/5.0196670 , Feb-2024
Articles in Peer-reviewed Journals	Topper B, Neumann A, Wilke SK, Youngman RE, Alrubkhi A, Weber R. "Comparing Pr3+ and Nd3+ for deactivating the Er3+: 4I13/2 level in lanthanum titanate glass." Optical Materials Express. 2024 May 1;14(5):1309-20. https://doi.org/10.1364/OME.517790 , May-2024